

Task 1

```
1 public class Student{
2     public string name;
3     public int age;
4     public string address;
5
6     public static string college = "ifc";
7 }
```

```
1 using System;
2
3 class Program
4 {
5     static void Main(string[] args)
6     {
7         Student student1 = new Student();
8         student1.name = "Ram";
9         student1.age = 10;
10        student1.address = "Kathmandu";
11
12        Student student2 = new Student();
13        student2.name = "Shyam";
14        student2.age = 11;
15        student2.address = "Bajhang";
16
17        Console.WriteLine("Student 1 Details:");
18        Console.WriteLine("Name: " + student1.name);
19        Console.WriteLine("Age: " + student1.age);
20        Console.WriteLine("Address: " + student1.address);
21
22        Console.WriteLine();
23
24        Console.WriteLine("Student 2 Details:");
25        Console.WriteLine("Name: " + student2.name);
26        Console.WriteLine("Age: " + student2.age);
27        Console.WriteLine("Address: " + student2.address);
28
29        Console.WriteLine();
30
31        Console.WriteLine("College (Static Field): " + Student.college);
32    }
33 }
```

Student 1 Details:

Name: Ram

Age: 10

Address: Kathmandu

Student 2 Details:

Name: Shyam

Age: 11

Address: Bajhang

College (Static Field): ifc

rakeshgurung@Rakeshs-MacBook-Air workshop3 %

Task 2

```
1  using System;
2
3  public class Calculator{
4
5      public void PrintWelcome(){
6          Console.WriteLine("Welcome to the Calculator");
7      }
8
9      public int Add(int num1, int num2){
10         return num1 + num2;
11     }
12
13     public int Multiply(int num1, int num2 = 1){
14         return num1 * num2;
15     }
16 }
17
```

```

1  using System;
2
3  class Program
4  {
5      static void Main(string[] args){
6
7          Calculator calc = new Calculator();
8
9          calc.PrintWelcome();
10
11         int sum = calc.Add(5, 10);
12
13         int product1 = calc.Multiply(4, 3);
14
15         int product2 = calc.Multiply(7);
16
17         Console.WriteLine("Addition Result: " + sum);
18         Console.WriteLine("Multiplication (4 * 3): " + product1);
19         Console.WriteLine("Multiplication (7 * default 1): " + product2);
20     }
21 }

```

```

● rakeshgurung@Rakeshs-MacBook-Air workshop3 % dotnet run
/Users/rakeshgurung/Documents/3rd Year/AD/workshop3/Student.cs(2,19)
  error CS0019: Operator '+' cannot be applied to operands of type 'int' and 'string'. Consider adding the null
  coalescing operator (??) to the expression.
/Users/rakeshgurung/Documents/3rd Year/AD/workshop3/Student.cs(4,19)
  error CS0019: Operator '+' cannot be applied to operands of type 'int' and 'string'. Consider adding the null
  coalescing operator (??) to the expression.
Welcome to the Calculator
Addition Result: 15
Multiplication (4 * 3): 12
Multiplication (7 * default 1): 7
○ rakeshgurung@Rakeshs-MacBook-Air workshop3 %

```

Task 3



```
1  using System;
2
3  public class ParameterDemo{
4
5      public void Increase(ref int number){
6          number += 10;
7      }
8
9      public void GetFullName(out string fullname){
10         fullname = "Rakesh Gurung";
11     }
12
13     public int SumAll(params int[] numbers){
14         int sum = 0;
15         foreach (int num in numbers){
16             sum += num;
17         }
18         return sum;
19     }
20 }
21
```



```
1  using System;
2
3  class Program
4  {
5      static void Main(string[] args){
6          ParameterDemo demo = new ParameterDemo();
7
8          int value = 20;
9          demo.Increase(ref value);
10         Console.WriteLine("Value after Increase: " + value);
11
12         demo.GetFullName(out string fullName);
13         Console.WriteLine("Full Name: " + fullName);
14
15         int sum = demo.SumAll(5, 10, 20, 15);
16         Console.WriteLine("Sum of numbers: " + sum);
17     }
18 }
19
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

- rakeshgurung@Rakeshs-MacBook-Air workshop3 % dotnet run
Value after Increase: 30
Full Name: Rakesh Gurung
Sum of numbers: 50
- rakeshgurung@Rakeshs-MacBook-Air workshop3 % □

Task 4



```
1  using System;
2
3  public class Player
4  {
5
6      public string playerName;
7      public int level;
8      public int health;
9
10
11     public Player(){
12         Console.WriteLine("Default constructor has been called");
13     }
14
15     public Player(string playerName, int level, int health){
16         this.playerName = playerName;
17         this.level = level;
18         this.health = health;
19     }
20 }
21
```



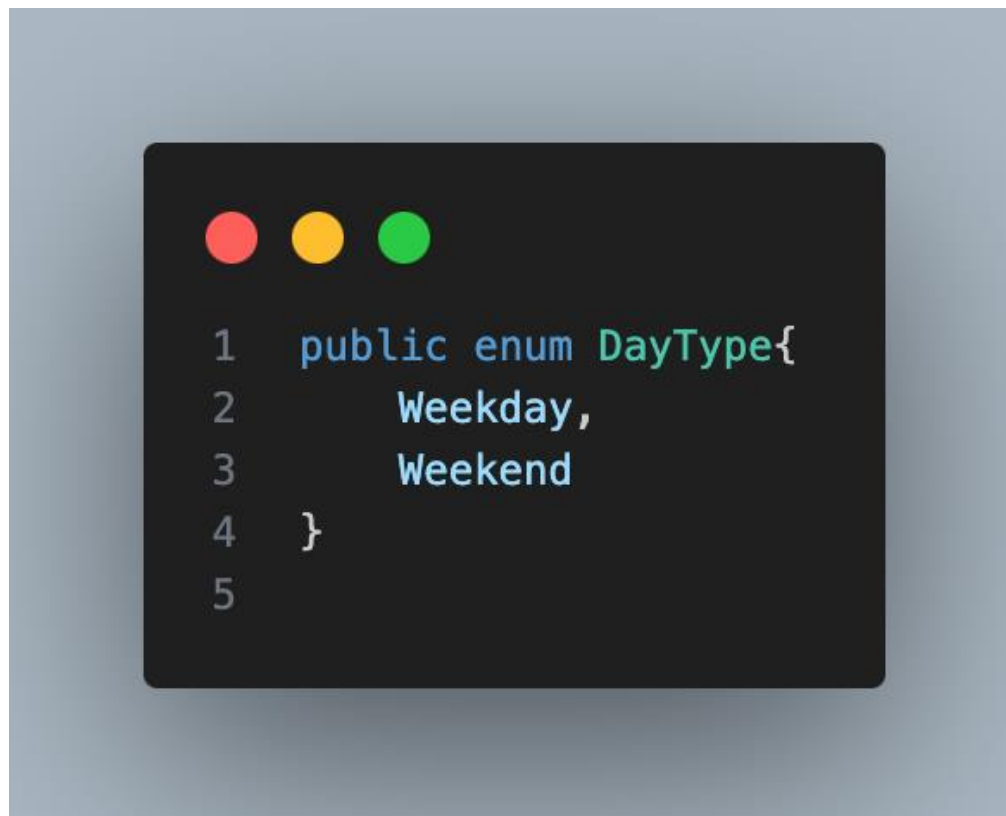
```
1  using System;
2
3  class Program
4  {
5      static void Main(string[] args){
6
7          Player p1 = new Player("KP Oli", 7, 250);
8
9          Console.WriteLine("Player 1 Details:");
10         Console.WriteLine("Name: " + p1.playerName);
11         Console.WriteLine("Level: " + p1.level);
12         Console.WriteLine("Health: " + p1.health);
13
14         Console.WriteLine();
15
16         Player p2 = new Player("Ramesh Lekhak", 5, 100);
17
18         Console.WriteLine("Player 2 Details:");
19         Console.WriteLine("Name: " + p2.playerName);
20         Console.WriteLine("Level: " + p2.level);
21         Console.WriteLine("Health: " + p2.health);
22     }
23 }
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● rakeshgurung@Rakeshs-MacBook-Air workshop3 % dotnet run
/Users/rakeshgurung/Documents/3rd Year/AD/workshop3/Player.cs(11,12): warning CS8618:
ust contain a non-null value when exiting constructor. Consider adding the 'required'
s nullable.
Player 1 Details:
Name: KP Oli
Level: 7
Health: 250

Player 2 Details:
Name: Ramesh Lekhak
Level: 5
Health: 100
○ rakeshgurung@Rakeshs-MacBook-Air workshop3 %
```

Task 5





```
1  using System;
2
3  class Program
4  {
5      static void Main(string[] args)
6      {
7          Console.Write("Enter a day: ");
8          string day = Console.ReadLine().Trim().ToLower();
9
10         days dayType;
11
12         if (day == "friday" || day == "saturday"){
13             dayType = DayType.Weekend;
14         }else{
15             dayType = DayType.Weekday;
16         }
17
18         Console.WriteLine("It is: " + dayType);
19     }
20 }
21
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS zsh

- rakeshgurung@Rakeshs-MacBook-Air workshop3 % dotnet run
Enter a day: sunday
It is: Weekday
- rakeshgurung@Rakeshs-MacBook-Air workshop3 % dotnet run
Enter a day: friday
It is: Weekend
- rakeshgurung@Rakeshs-MacBook-Air workshop3 %

Task 6

```
1  using System;
2
3
4  class Program{
5      static void Main(string[] args){
6
7          // Prompt for marks
8          Console.Write("Enter marks: ");
9          string marksInput = Console.ReadLine();
10
11         // Prompt for total
12         Console.Write("Enter total: ");
13         string totalInput = Console.ReadLine();
14
15         // Parse inputs using TryParse
16         bool marksParsed = int.TryParse(marksInput, out int marks);
17         bool totalParsed = int.TryParse(totalInput, out int total);
18
19         // Check if parsing succeeded
20         if (!marksParsed || !totalParsed)
21         {
22             Console.WriteLine("Invalid input. Please enter valid integers.");
23             return;
24         }
25
26         // Corrected calculation: Cast marks to double for floating-point division
27         double percentage = (double)marks / total * 100;
28
29         // Print the result
30         Console.WriteLine($"Percentage: {percentage}%");
31     }
32 }
```

- rakeshgurung@Rakeshs-MacBook-Air workshop3 % dotnet run
Enter marks: 87
Enter total: 100
Percentage: 87%
- rakeshgurung@Rakeshs-MacBook-Air workshop3 %